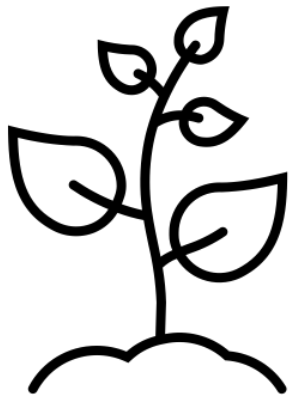




The Hurt Dirt

An analysis of the external contributors of
soil pollution

introduction

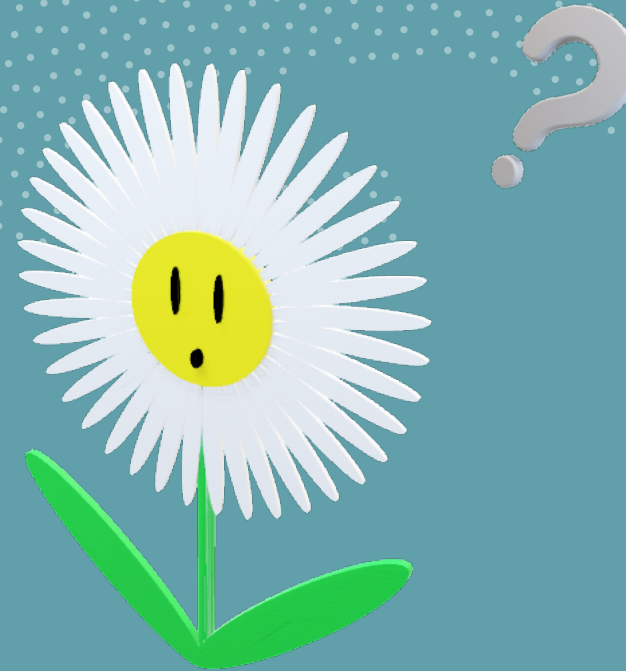


Despite “dirt” being a derogatory statement, it is an essential life sustaining component of Earth.

This analysis explores the effects of external pollutants on soil health.

Understanding these factors environmental effects are important in maintaining sustainability efforts.

Problem Statement



Can we predict the influence of soil pollution based on environmental factors?

Measuring the dataset's variables and their interactions can produce information about conditions that lead to increasing pollution and reduction methodologies. Data can show the common diseases, frequent health impacts, and influences including rainfall, soil pollutant concentrations, and nearby industries.

The Significance of it All



The world depends on healthy soil. Polluted soils pose risks to every living organism on Earth. It is worth investigating the effects that environmental pollution has on fertile dirt. Natural land is responsible for both the world's agriculture. Ongoing soil pollution causes prolonged environmental damage cause unsustainable crop growth. These conditions can lead to limited food production and eventually, famine. Studying this phenomenon can help build towards discovering sound solutions for all crops requiring healthier soil.

Dataset Summary

Dataset	<i>Soil Pollution and Associated Health Impacts</i>
Source	https://www.kaggle.com/datasets/khushikyad001/soil-pollution-and-associated-health-impacts/data
Size	3,000 synthetic records across 24 variables
Data Types	Mix of categorical, geospatial, nominal, continuous, numerical, ordinal, and qualitative binary data
Relevance	The dataset provides a robust set of variables that can be combined and measured to explore the impacts that polluted soils have on the environment

Types of Visualization



Bar chart

A bar chart can show us the number of occurrences of different types of soil pollutants. We can then identify which pollutants are the most common.

Pie Chart

A pie chart will show us the distributions of the kinds of diseases that occur within the data set. We will see which diseases occur both the most and least.

Correlation Matrix

Our dataset has six variables that handle numerical values. Among these values, it would be fascinating to draw correlations between each to see if there are any linear relationships between the two.

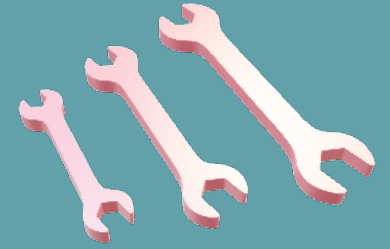
Box Plot (Mixed: Continuous vs. Categorical)

Comparing the distribution of the types of pollutants and their concentrations with nearby industries can give us a sense of which industries contribute the strongest pollutant concentrations for soil.

Line Chart

The line chart can show us pollutant concentrations by pollutant type over time within specific regions and countries. In this case, we will look at the United States of America.

TOOLS AND TECHNIQUES



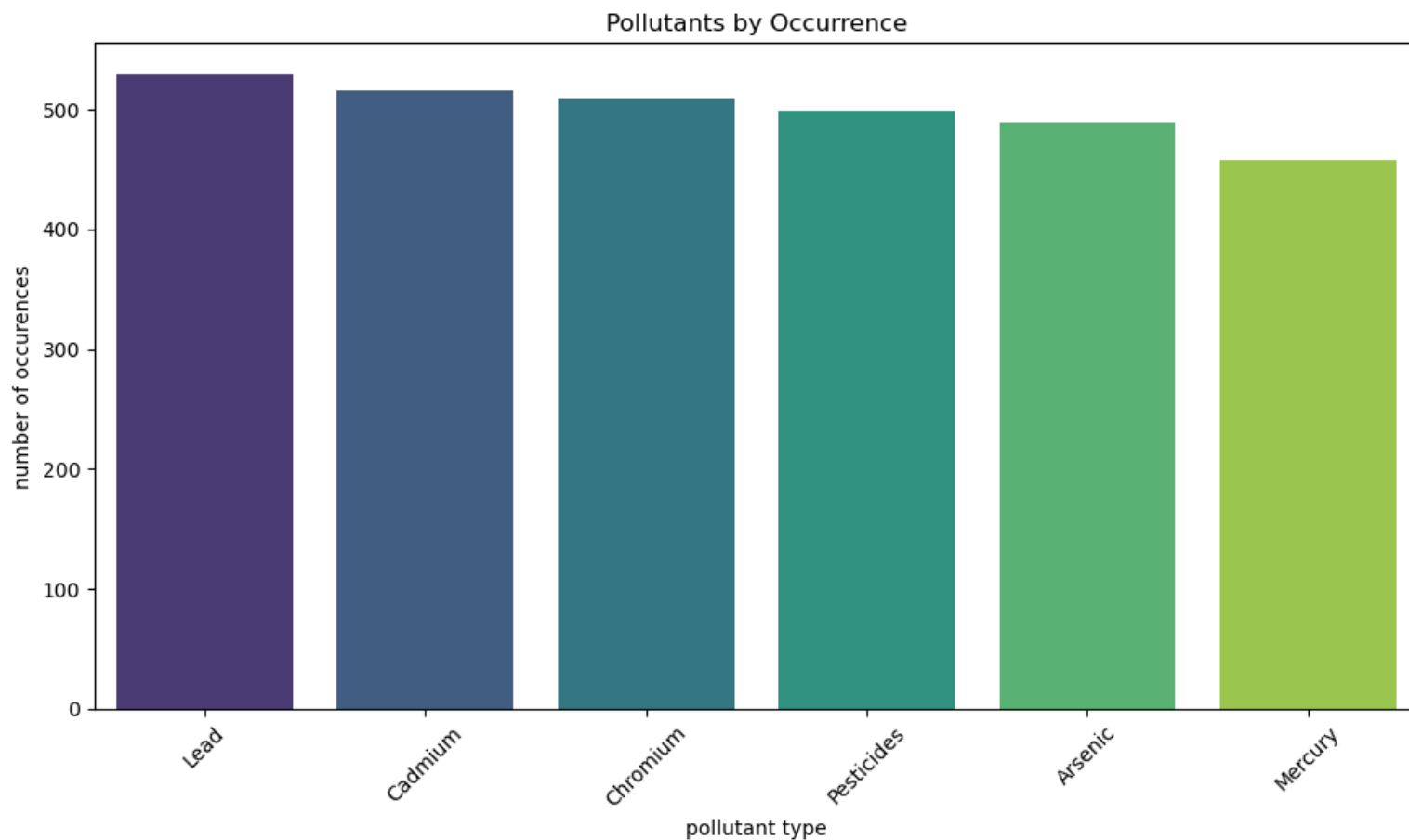
Python Libraries

- `matplotlib` and `seaborn` were used for the bar chart, pie chart, and the correlogram.
- `Plotly` was used for the interactive line chart and boxplot graph.

Data Processing

- Customized colors, rotating axis titles, and placing keys to increase readability for users.
- Cleaning irrelevant data to provide more accurate chart readings.

BAR CHART

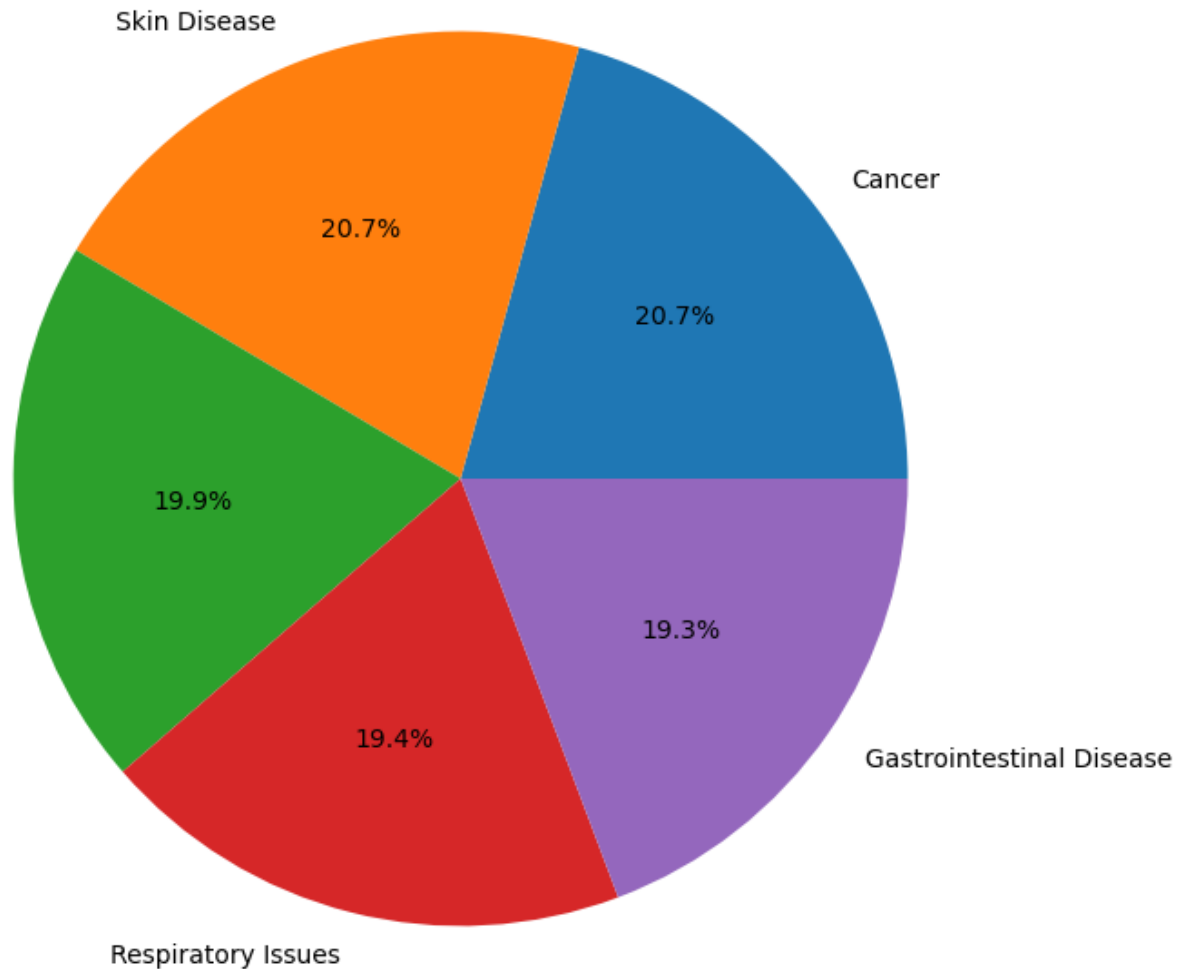


Findings

All pollutant types are relatively high, occurring more than 400 times each within the dataset. The leading soil pollutant is lead, with cadmium, chromium, pesticides, and arsenic trailing closely behind. Mercury falls off as the least occurring of the pollutants but still has a high occurrence that's not that different in size to the rest.

PIE CHART

Distributions by Disease

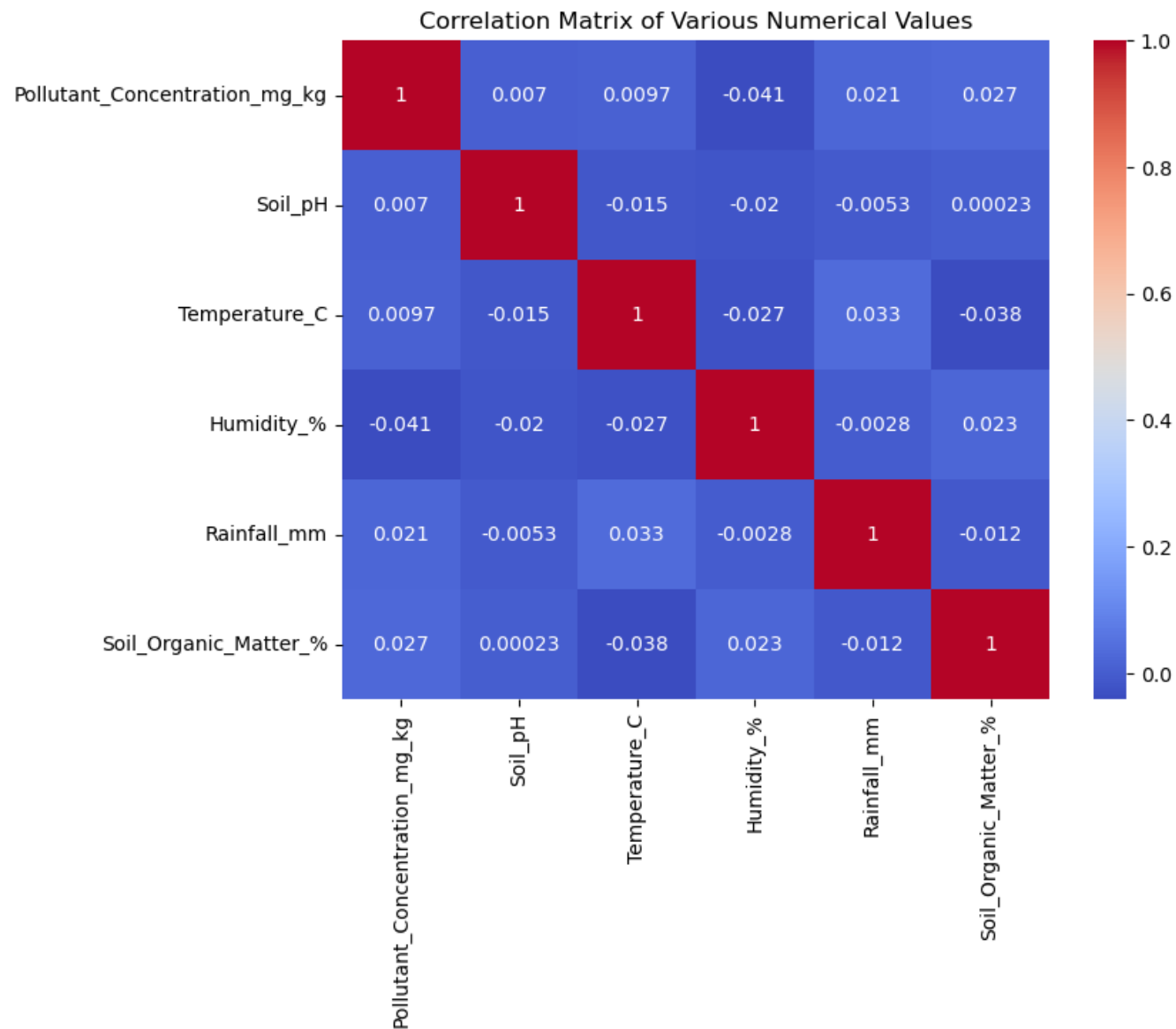


Findings

Very similar do we find in scope and proportion the types of diseases that occur from polluted soil. Both skin disease and cancer tie and rank among the highest of all diseases.

Gastrointestinal disease occurs less, but not more than one percent for the third and fourth most common disease occurrences.

CORRELATION MATRIX



Findings

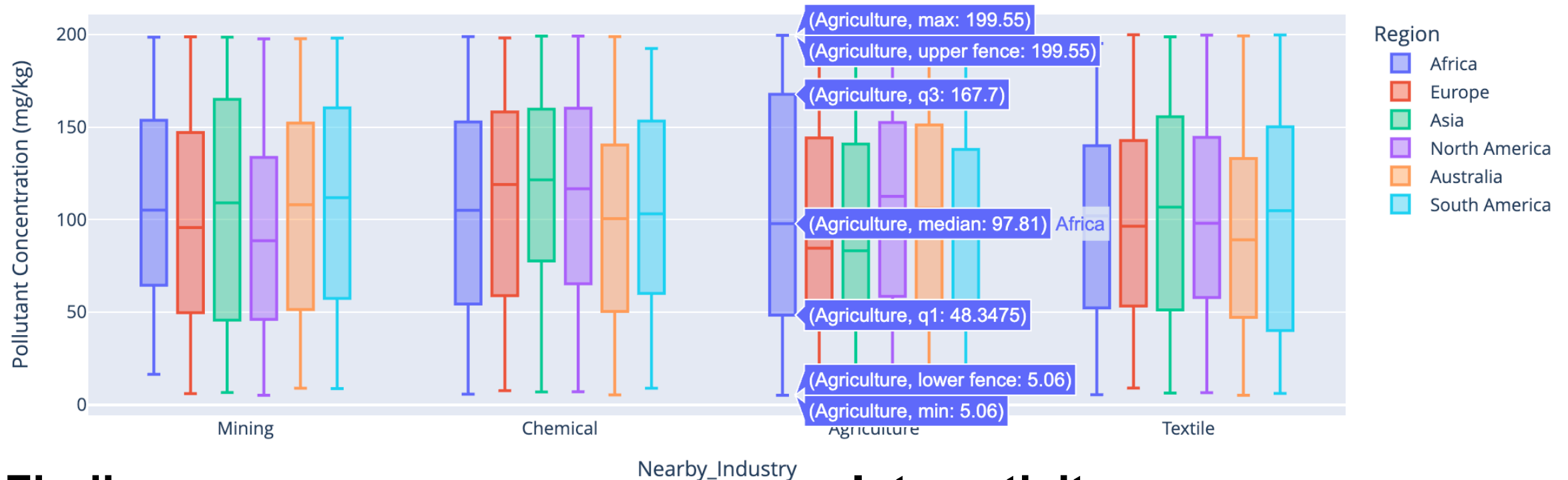
The data is suspected to be entirely synthetic. No correlations can be found between various numerical values like soil pH levels and rain fall (which, in reality, is assumed to have a stronger correlation due to acidity.)

Despite the limitation of this data, we can state that there are no significant correlations between each numerically represented variable. This indicates that these variables show little to no effects between their relationships.

BOX PLOT

interactive

Pollutant Concentration vs Nearby Industry



Findings

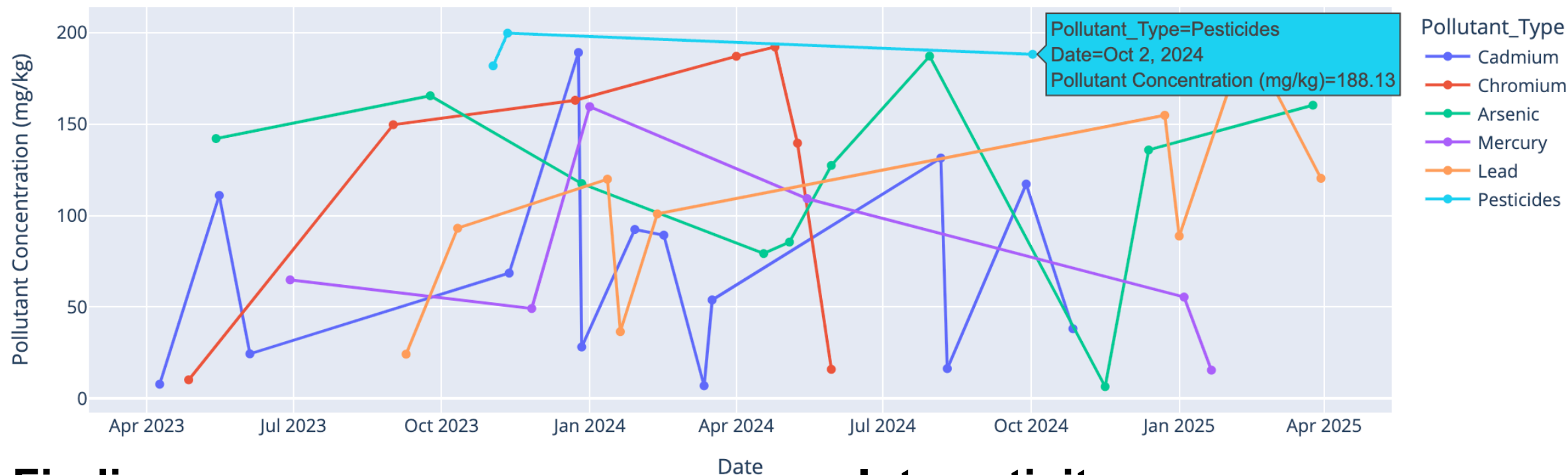
By far the highest median of soil pollutant concentration lies within the chemical industry, no matter the region. Some medians show jumps between regions, perhaps indicating that more of these areas have more specific types of industries.

Interactivity

Users can hover over each bar graph representing each region (continent) to get a popup with more details. These details show numerical ranges including, but not limited to, the interquartile range and values both minimum and max.

LINE CHART interactive

Pollutant Concentration Over Time in the USA (North America)



Findings

The sporadic results of pollutant concentrations show the data's true randomness of its synthetic nature. One insight we can pull is that lead shows consistent signs of higher concentrations over the years with only a couple of dips.

Interactivity

The interactivity aspect for the line chart allows users to hover over certain points in time to show a popup. This popup shows the pollutant type, date reported, and the amount of pollutant concentration.



KEY FINDINGS SUMMARY

In the United States, lead is the leading soil pollutant. It shows an annual sustained quantitative increase. Lead is the biggest threat of pollutants for soil health.

Chemical industries worldwide also contribute the most towards soil pollution. No matter the continent, the most damaging industry for soil health is the chemical industry.

There are no significant connections between naturally occurring environmental conditions like rainfall, soil pH levels, temperatures, and other variables tied to pollutant concentrations. This suggests that industries alone contribute the most soil pollution, with no other variable causing significant alterations.

*** = Synthetic data** points unfortunately skew the data where finding patterns becomes difficult. Data from real world settings presumes clearer and less sporadic results than the ones presented in this project.



thank you!

The End.

Thank you for viewing this presentation. 😊